Kerr Landau-Siegel proof evaluation, Step 2: line-by-line audit

Document evaluated: Bryce Kerr, "A simpler proof that there are no Landau–Siegel zeros," arXiv:2610.21099v1 [math.NT], 15 October 2026, 6 pages. This is the copy Arnoldo attached on 18 October 2026.

Built on: "Step 1 - Kerr Landau-Siegel proof evaluation Step 1 18 Oct 2026 1708_v1.md". Item numbers F1 to F12 refer to its foundation inventory.

Sources consulted in this step: Davenport, *Multiplicative Number Theory*, 2nd edition, pages 88 and 89 (images Arnoldo supplied). I used them only for the derivation of Lemma 1 in Part C. Masser and Milne are compared against Kerr's statements in Step 4, not here.

Standard applied: a proof must carry the reader from step to step with the reader supplying nothing. Completeness is judged only by what is on the page. Every unexecuted step is a break. A false step (a statement that is wrong as written) is recorded separately, because it is worse than a gap.

Labels used in this file

Label	Meaning
Executed	The step is on the page and correct.
Break	The step is not executed on the page. Each one gets a severity. *Routine:* an expert supplies it in a line or two from the stated or standard foundation. *Substantive:* real work, a case, or a choice is missing.
False step	The statement is wrong as written.
Notation defect	The mathematics is unaffected, but the reader has to resolve an ambiguity.

Every derivation that fills a break is in **Part C** and is labeled as mine. None of it is credited to Kerr. In Parts A and B, "true" means "true by the derivation in Part C", not "true on the page".

Part A. Audit in page order

A.1 Page 1: Abstract, Introduction and Theorem 1

#	Location	Finding	Label
A1	Whole paper	The paper never states its foundation up front. Davenport, Milne and Masser are cited where they are used. Mertens' estimate, the Chinese remainder theorem and the pigeonhole principle are named but not stated. Euler's criterion, the Chebyshev bound, the properties of the field norm, and the linear algebra used in Lemmas 2 and 7 appear nowhere (Step 1, F3 and F7 to F12).	**Break B1** (substantive: the reader must reconstruct the foundation)
A2	Introduction: "OpenAI [3] recently established that Siegel zeros do not exist"	This is an outside claim. The proof does not use it, since Kerr's argument is self-contained apart from F1 to F12. It is not a proof step, so it is not counted. It is flagged for Step 5.	(not counted)
A3	Theorem 1	The statement covers every fundamental discriminant q, including small ones. The proof by contradiction covers only $\|q\|$ large, "sufficiently large" in terms of γ and ε. No sentence handles the finitely many remaining q, and the constant c is never named. Closing this needs two facts that appear nowhere in the paper: $L(1, \chi) \neq 0$, and the zeros of $L(s, \chi)$ are isolated (Part C.11).	**Break B2** (substantive)
A4	Theorem 1 and Lemma 1	Davenport's inequality (2) applies to *primitive* characters, with log q where q is the conductor. Kerr needs $\chi = (q/\cdot)$ to be primitive of conductor $\|q\|$. The Theorem's hypotheses say "primitive". That the conductor	**Break B3** (routine)

		equals |q| for a fundamental discriminant is used silently. This is a new foundation item, F13, not listed in Step 1.	

A.2 Page 1: Lemma 1 (analytic input)

#	Location	Finding	Label
A5	"a straightforward consequence of [2, Equation (2), p. 89]"	This is an assignment, not a derivation. The derivation (Part C.1) also needs the *unnumbered* display at the top of Davenport p. 89, the bound $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) + c_1$ for $1 < \sigma < 2$. Kerr does not cite it; this is new item F15. The derivation also needs a choice of σ (σ = 1 + 1/log X), the positivity of 1 + χ(n), and the lower bound p^{−σ} ≥ e^{−1}/p for p ≤ X. None of these is on the page. The statement is **true** (Part C.1).	**Break B4** (substantive: the lemma that carries the argument is not proved on the page)

A.3 Page 1: Number fields

#	Location	Finding	Label
A6	"The field K has degree four"	**False as written for q = 8**, which is a fundamental discriminant. Then √8 = 2√2, so K = Q(√2) has degree 2. Every other fundamental discriminant gives degree 4, because the only q with Q(√q) = Q(√2) is q = 8. The paper states no restriction.	**False step FS1**
A7	"O has Z-basis 1, √q, √2, √(2q)"	This also fails for q = 8, where √(2q) = 4 and √q = 2√2 make the four elements dependent; it is part of FS1. For every other q it is true, but not shown. It needs [K : Q] = 4, and the fact that the Z-span of these four elements is closed under multiplication (Part C.2).	**Break B5** (routine), plus FS1
A8	Embeddings, σ and τ	The convention √(2q) := √2 · √q is never stated. Neither is the branch of √q for q < 0, nor the fixed embedding K ↪ C needed in the proof of Theorem 1 to place (θ_n, σθ_n, στθ_n) and $\gamma_1, \ldots, \gamma_4$ in C^3. The existence of σ and τ (K/Q is Galois with group {1, σ, τ, στ}) is asserted ("determined by"), not justified.	**Break B6** (routine)
A9	‖α‖ := max_t |α^{(t)}|	Executed (a definition).	Executed

A.4 Page 2: Equation (2) and the Frobenius congruences

#	Location	Finding	Label
A10	"For p ∤ 2q the action of Frobenius modulo p, see [4, §8, Example 8.19], gives (√q)^p ≡ χ(p)√q, (√2)^p ≡ (2/p)√2 (mod pO)"	This is an assignment. Milne's example identifies the Frobenius *element* of Gal(Q(√d)/Q) with the Legendre symbol (d/p). Turning that into a congruence modulo pO, in the ring O = Z[√q, √2] (which is not the ring of integers of K in general), takes further steps that are not on the page. The step from Kronecker χ(p) to Legendre (q/p) for odd p is also silent (F3). Both congruences are **true**, and follow in one line from Euler's criterion, with no Frobenius theory needed (Part C.3).	**Break B7** (routine)
A11	"Hence, if χ(p) = −1 then θ_n^p ≡ σθ_n … (3), θ_n^p ≡ στθ_n … (4)"	The "Hence" is not executed. The step needs: the multinomial expansion modulo p in the commutative ring O; n^p ≡ n (mod p) for integers n (Fermat); (√(2q))^p	**Break B8** (routine)

		= (√2)^p(√q)^p under the convention of A8; and the sign bookkeeping against σ and στ. **True** (Part C.3).	

A.5 Page 2: Polynomials and Lemma 2

#	Location	Finding	Label
A12	Definition of L_D (5), (6)	Executed. The exponents i, j, k ≥ 0 are implicit.	Executed
A13	Lemma 2, proof: "the change of basis matrix from (5) to (7) is a triangular matrix with diagonal entries one"	Three things are missing. **(a)** Each G_{i,j,k} lies in L_D: expanding (x₁^p − x_ℓ)^e gives monomials of weighted degree i − t(p − 8) + 8(j + k) ≤ D. This is the only place p > 8 is used, and the paper never says so. **(b)** No ordering of monomials is given under which the matrix is triangular. Ordering by the x₁ exponent works. **(c)** An integer unitriangular change of basis is invertible over Z, hence over O. **True** (Part C.4).	**Break B9** (routine)

A.6 Page 2: Lemma 3

#	Location	Finding	Label
A14	Definition of ‖f‖_B	Executed (a definition). The weight is B^{i+j+k}, the *total* degree, not the weighted degree. This is consistent with its later use.	Executed
A15	"These choices satisfy ‖c_{ijk}‖B^{i+j+k} ≤ H"	Not shown. It needs ‖u₁ + u₂√q + u₃√2 + u₄√(2q)‖ ≤ U(1 + √2)(1 + √|q|), and (1 + √2)(1 + √|q|) ≤ 8√|q| for |q| ≥ 3. **True** (Part C.5).	**Break B10** (routine)
A16	"there are more than (H/(8√|q|B^{i+j+k}))⁴ choices for each coefficient the result follows"	The count (U + 1)⁴ > x⁴ with U = ⌊x⌋ is executed. Two facts are silent: distinct (u₁, …, u₄) give distinct coefficients (the Z-basis of A7, which fails for q = 8), and distinct coefficient tuples give distinct f. Lemma 3 inherits FS1 for q = 8.	**Break B11** (routine)

A.7 Page 3: Lemma 4

#	Location	Finding	Label
A17	Statement	Executed as a statement. The definitions of r, E_p and P are clear on the rendered page: E_p = Σ⌊i/p⌋.	Executed
A18	"By Lemma 2 each f ∈ L_D can be expressed in the form Σ c_{i,j,k} G_{i,j,k}"	The symbol c_{i,j,k} is reused for the coordinates in the G basis, which differ from the monomial coefficients of (6).	**Notation defect ND1**
A19	"By (3) and (4), … G_{ijk}(θ_n, σθ_n, στθ_n) ≡ 0 (mod p^{⌊i/p⌋}O)"	Not shown. The step uses three facts: with ℓ chosen as stated, x₁^p − x_ℓ evaluates into pO by (3) or (4); its e-th power, e = ⌊i/p⌋, lies in p^eO; and the remaining factors lie in O, because θ_n ∈ O and σ, τ preserve O. **True** (Part C.6).	**Break B12** (routine)
A20	"Consequently, the values of c_{i,j,k} mod p^{k_p−⌊i/p⌋}O determine the vector"	Not shown. The exponent k_p − ⌊i/p⌋ must be non-negative, which follows from ⌊i/p⌋ ≤ ⌊D/p⌋ ≤ k_p because i ≤ D. That is the purpose of the hypothesis k_p ≥ ⌊D/p⌋, and the paper never says so. **True.**	**Break B13** (routine)
A21	"= Π p^{4(k_p−⌊i/p⌋)} = p^{4k_p r − 4E_p}"	The first equality uses |O/mO| = m⁴, because O is free of rank 4 over Z. That fact is silent and fails for q = 8. The second equality is executed.	**Break B14** (routine)
A22	"the Chinese remainder theorem	The Chinese remainder theorem is	**Break B15** (routine)

	gives"	named but not stated, either in the form O/PO ≅ Π_p O/p^{k_p}O or as the injection that bounds the joint count by the product of the counts. **True.**	

A.8 Pages 3 and 4: Lemma 5

#	Location	Finding	Label
A23	Statement: "For sufficiently large N"	The threshold is never made explicit, and the paper never says whether it depends on γ. Part C.8 shows that it can be taken absolute. Absorbed into B21.	(see B21)
A24	"By Lemma 3, the number … exceeds (P/(32r√|q|))^{4r} B^{−4W}"	Executed: the product of r factors (H/(8√|q|))⁴ B^{−4(i+j+k)} with H = P/(4r).	Executed
A25	"If (11) then by Lemma 4 there exist distinct f_1, f_2"	The algebra that turns "count of polynomials ≥ count of residue vectors" into (11), with P cancelling, is not shown. **True** (Part C.7).	**Break B16** (routine)
A26	"‖θ_n‖, ‖σθ_n‖, ‖στθ_n‖ ≤ B"	Not shown. It needs ‖θ_n‖ ≤ (N − 1)(1 + √2)(1 + √|q|) < 8N√|q|, and the fact that σ and τ permute the embeddings, so ‖σθ‖ = ‖θ‖. **True.**	**Break B17** (routine)
A27	"‖F(θ_n, σθ_n, στθ_n)‖ ≤ 2rH = P/2"	Not shown. It uses ‖F‖_B ≤ 2H (triangle inequality for each coefficient), then the triangle inequality across r monomials in each embedding, then A26. **True.**	**Break B18** (routine)
A28	"its norm from K to Q would be a nonzero integer divisible by P⁴, while the absolute value of this norm is at most (P/2)⁴"	Three facts are silent: N_{K/Q} maps O {0} to nonzero rational integers, because O lies in the ring of integers; N(Pβ) = P⁴N(β); and |N(α)| ≤ ‖α‖⁴. **True.**	**Break B19** (routine)
A29	"Uniformly for p ≤ Q, we have r = D³/384 + O(D²), W = 5D⁴/6144 + O(D³), E_p = D⁴/(1536p) + O(D³)"	All three asymptotics are stated without derivation, and so is the uniformity in p. All three leading constants are **correct** by hand (Part C.8). Step 3 will confirm them by exact computation.	**Break B20** (substantive: these numbers produce the 5/4 in the margin chain)
A30	"Σ E_p log p = (D⁴/1536) Σ (log p)/p + O(D³Q)"	The error term O(D³Q) uses the Chebyshev bound Σ_{p≤Q} log p ≪ Q silently (F11).	Part of B21
A31	"from which (11) follows"	Not executed. The step needs: (9) gives the main term (13/10)(D⁴/1536) log B; W log B has main term (5/4)(D⁴/1536) log B; the margin of 1/20 must dominate O(D³Q) = O(D⁴/(log N)²), O(D³ log B) and r log(32r√|q|) = O(D³ log N); and the threshold on N must be independent of γ, because γ is fixed before |q|. **True** (Part C.8).	**Break B21** (substantive)

A.9 Page 4: Lemma 6 (Masser)

#	Location	Finding	Label
A32	Definition of μ(Γ)	Executed as a definition. "Lattice" is used for any finitely generated subgroup of C^n, which need not be discrete.	**Notation defect ND2**
A33	Lemma 6, attributed to Masser [5]	This is an import, with no theorem number or page. In the source, Masser's Theorem 1 (p. 82) **assumes $\gamma_1, \ldots, \gamma_l$ form a basis** of Γ, and requires D ≥ 1. Kerr's Lemma 6 allows *any* generators $\gamma_1, \ldots, \gamma_s$. As stated, it says more than the source does. The paper's application needs only the basis case: $\gamma_1, \ldots, \gamma_4$ form a basis of Γ, because Lemma 7 shows Γ has rank 4. That is never pointed out.	**Break B22** (substantive: the citation does not cover the statement as written)

		Whether Kerr's μ is the same as Masser's μ is not shown. Step 4 will settle both questions against the original.	

A.10 Page 5: Lemma 7

#	Location	Finding	Label
A34	"The hypothesis makes π injective on Z^4, so Γ has rank 4"	Not shown. The argument: since π is surjective, ker π has dimension 1. A nonzero $x \in Z^4 \cap \ker \pi$ would give $\ker \pi = Cx$, a proper subspace spanned by a rational vector, which the hypothesis forbids. **True.**	**Break B23** (routine)
A35	"Since W is proper"	Not shown. $W \subseteq \pi^{-1}(L)$, and $\dim \pi^{-1}(L) = \dim L + 1 \le 3$. **True.**	**Break B24** (routine)
A36	"our assumption implies $\ker \pi \cap W = \{0\}$"	Executed in outline. It needs $\dim \ker \pi = 1$, which is also missing in A34.	Executed (inherits B23)
A37	"$\mathrm{rank}_Z(\Gamma \cap L) = \dim_C W = \dim_C \pi(W) \le \dim_C L$"	Three facts are silent: $\Gamma \cap L = \pi(\pi^{-1}(L) \cap Z^4)$; for a subgroup of Z^4, the Z-rank equals the complex dimension of its span (integer vectors independent over Q are independent over C); and $\pi(W) \subseteq L$. **True.**	**Break B25** (routine)
A38	"Thus $\mu(\Gamma) \ge \min_{0 \le h < 3} (4 - h)/(3 - h) = 4/3$"	Executed except for one silent fact: $\mathrm{rank}(\Gamma/(\Gamma \cap L)) = \mathrm{rank}\,\Gamma - \mathrm{rank}(\Gamma \cap L)$. The minimum is checked: 4/3, 3/2 and 2 for h = 0, 1, 2.	Executed (routine silent fact folded into B25)

A.11 Pages 5 and 6: Proof of Theorem 1

#	Location	Finding	Label
A39	"Suppose that $(1 - \beta) \log\|q\| < \varepsilon$. Fix a sufficiently large positive integer γ …" and later "provided γ is sufficiently large, ε sufficiently small in terms of γ, and |q| sufficiently large"	The order of quantifiers is inverted on the page: ε appears before γ, but must be chosen after it. The correct order (γ, then ε, then q_0, then the contradiction hypothesis) is never written.	**Break B26** (substantive: this is the logical form of the theorem)
A40	"Mertens' estimate and Lemma 1 give $\Sigma_{p \in S} (\log p)/p \ge \log Q - O(\log\|q\| + (1 - \beta)(\log Q)^2)$"	Not executed. Mertens' estimate is not stated. The primes excluded from S ($p \le 8$, and $p \mid q$ where $\chi(p) = 0$) are not accounted for. The application of Lemma 1 needs $X = Q \ge 3$. **True** (Part C.9).	**Break B27** (routine)
A41	"Since $\log Q = (4/3) \log N + O(\log \log N)$ and $\log B = \log N + \tfrac{1}{2} \log\|q\| + O(1)$, we have (9)"	The two asymptotic statements are true and nearly immediate. The deduction of (9) is **not executed.** It has a margin of only $4/3 - 13/10 = 1/30$ in the coefficient of log N. It requires bounding $(1 - \beta)(\log Q)^2$ by about $(16/9)\varepsilon\gamma \log N$, and bounding the $\tfrac{1}{2} \log\|q\|$ and $O(\log\|q\|)$ terms by multiples of $(1/\gamma) \log N$. This is the decisive inequality of the proof. **True** (Part C.9).	**Break B28** (substantive)
A42	"Lemma 5 supplies a nonzero $F \in L_D$ satisfying (12)"	Executed, given (9) and the "N sufficiently large" condition, which follows from |q| large.	Executed
A43	"We apply Lemma 6 with $\gamma_1 = (1, 1, 1)$, …" and "$(\theta_n, \sigma\theta_n, \sigma\tau\theta_n) = n_1\gamma_1 + \cdots + n_4\gamma_4$"	The identity checks correctly by hand against the definitions of σ and στ (Part C.10). The silent part: F has coefficients in $O \subset K$, and becomes a nonzero complex polynomial with the same degree only after the fixed embedding of A8 is applied.	**Break B29** (routine; same root as B6)
A44	The integer γ and the vectors $\gamma_1, \ldots, \gamma_4$	The symbol γ is used for both.	**Notation defect ND3**
A45	"$\det(\gamma_1, \gamma_2, \gamma_3) = 4\sqrt{(2q)} \ne 0$, the map π is surjective and its kernel is spanned by $v = (\sqrt{(2q)}, \sqrt{2}, -\sqrt{q}, -1)$"	The determinant value is **correct** by hand. That $\pi(v) = 0$, and that the kernel has dimension exactly 1 (rank 3), are not shown. **True** (Part C.10).	**Break B30** (routine)

A46	"The coordinates of v are linearly independent over Q, thus ker π lies in no proper complex subspace spanned by rational vectors"	Two problems. **(a)** The independence is asserted. It is equivalent to [K : Q] = 4, so it is **false for q = 8** (FS1 again), where √(2q) = 4 and √q = 2√2. **(b)** The "thus" needs the fact that a proper subspace spanned by rational vectors lies in a hyperplane $\Sigma\, a_i x_i = 0$ with $a \in Q^4 \setminus \{0\}$. That fact is silent (F8).	**Break B31** (routine), plus FS1
A47	"Lemma 6, with M = N − 1, now gives deg F ≥ ((N − 1)/3)^{4/3} > D contradicting that F ∈ L_D"	Four steps are silent. (i) (M/3)^μ ≥ (M/3)^{4/3} needs M/3 ≥ 1, which holds because N ≥ 4. (ii) D < ((N − 1)/3)^{4/3} follows from the definition of D. (iii) F ∈ L_D gives total degree i + j + k ≤ i + 8(j + k) ≤ D. (iv) Masser's basis hypothesis holds here (B22).	**Break B32** (routine)
A48	End of proof	The proof ends with a contradiction for large |q| only. Small |q| is B2.	(see B2)

Part B. Summary

B.1 False steps

#	Statement	Consequence	Repair
FS1	"The field K has degree four," with the Z-basis claim (A7), the Lemma 3 count (A16), $\|O/mO\| = m^4$ (A21), and the Q-independence of the coordinates of v (A46). All are stated for every fundamental discriminant q.	All are false for q = 8. No other fundamental discriminant is affected.	Assume |q| > 8 throughout. The proof needs |q| large anyway, so the repair costs nothing, but the page never states it.

No other false step was found. Finding none is not proof that none exists. Lemma 6 as written (any generators rather than a basis) is held open for Step 4. It is classed as a break, not a false step, because I have not shown it false. It only goes beyond its cited source.

B.2 Breaks

32 breaks in total: 8 substantive and 24 routine.

Substantive (8):

#	Where	What is missing
B1	Whole paper	Foundation not stated up front
B2	Theorem 1	Small |q| not handled; constant c never named
B4	Lemma 1	Derivation from Davenport absent
B20	Lemma 5	Asymptotics for r, W, E_p not derived
B21	Lemma 5	"(9) implies (11)" not executed
B22	Lemma 6	Stated more generally than its source; basis hypothesis dropped
B26	Theorem 1 proof	Quantifier order inverted
B28	Theorem 1 proof	The decisive inequality (margin 1/30) not executed

Routine (24): B3, B5 to B19, B23 to B25, B27, and B29 to B32.

B.3 Notation defects (3)

- **ND1:** c_{i,j,k} is reused for the coordinates in the G basis.
- **ND2:** "lattice" is used for a subgroup that need not be discrete.
- **ND3:** γ is used both as an integer and for the vectors $\gamma_1, \ldots, \gamma_4$.

B.4 Additions to the Step 1 foundation inventory

Step 1 v1 is left unchanged. These items are added here:

- **F13:** for a fundamental discriminant q, χ = (q/·) is a primitive character of conductor |q| (B3).
- **F14:** L(1, χ) ≠ 0, and the zeros of the entire function L(s, χ) are isolated. Needed only for the small-|q| case the paper

omits (B2).
- **F15:** Davenport p. 89, the unnumbered display $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) + c_1$ for $1 < \sigma < 2$. Needed for Lemma 1 but not cited (B4).

B.5 Verdict of this step, on the page only

Above the foundation, every step I could check is **true**, apart from the $q = 8$ statements, which are harmless once $|q|$ is assumed large. The paper is not complete under our standard. The four places that carry the argument (Lemma 1; the asymptotics and "(9) implies (11)"; the decisive inequality; the small-$|q|$ case) are all assigned to the reader. The import that carries the argument, Lemma 6, is stated more broadly than its source. Every break is fillable. Part C fills all of them except B22, which goes to Step 4 against Masser's original, and B1, which is a matter of presentation.

Part C. Derivations executed by me (not credited to Kerr)

These fill the breaks above. They are mine. Arithmetic marked "by hand" will be confirmed by machine in Step 3.

C.1 Lemma 1 from Davenport p. 89 (fills B4)

Write $\chi = (q/\cdot)$, which is primitive of conductor $|q| \geq 3$ (F13). Take real $s = \sigma \in (1, 2)$, so $t = 0$, and Davenport's (3) gives $\mathscr{L} = \log|q| + \log 2$. Davenport's (2) says $-L'/L(\sigma, \chi) < c_2\mathscr{L} - \Sigma_\rho \operatorname{Re} 1/(\sigma - \rho)$. Each term $\operatorname{Re} 1/(\sigma-\rho) = (\sigma - \operatorname{Re}\rho)/|\sigma-\rho|^2$ is ≥ 0 (as Davenport notes on the same page). Discarding every term except $\rho = \beta$ (a nontrivial zero, since $\beta \in (0, 1)$):

$$-L'/L(\sigma, \chi) < c_2 \log(2|q|) - 1/(\sigma - \beta).$$

Davenport's unnumbered display on p. 89 gives $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) + c_1$ (F15). Add the two, and use $-L'/L = \Sigma \chi(n)\Lambda(n)n^{-\sigma}$ and $-\zeta'/\zeta = \Sigma \Lambda(n)n^{-\sigma}$:

$$\Sigma_n (1 + \chi(n)) \Lambda(n)\, n^{-\sigma} < c_1 + c_2 \log(2|q|) + 1/(\sigma-1) - 1/(\sigma-\beta).$$

The last two terms equal $(1-\beta)/((\sigma-1)(\sigma-\beta)) \leq (1-\beta)/(\sigma-1)^2$, because $\sigma - \beta > \sigma - 1$. Every term on the left is ≥ 0, since χ is real and so $1 + \chi(n) \in \{0, 1, 2\}$. Keep only the primes $p \leq X$ with $\chi(p) = 1$, each of which contributes $2 \log p \cdot p^{-\sigma}$. Now take $\sigma = 1 + 1/\log X$, which lies in $(1, 2)$ because $X \geq 3 > e$. For $p \leq X$, $p^{-\sigma} = p^{-1} \cdot p^{-1/\log X} \geq e^{-1}/p$. Therefore

$$(2/e) \Sigma_{p\leq X,\, \chi(p)=1} (\log p)/p < c_1 + c_2 \log(2|q|) + (1-\beta)(\log X)^2,$$

and the right side is $\ll \log|q| + (1-\beta)(\log X)^2$, because $|q| \geq 3$. ∎

C.2 The ring O (fills B5, for $|q| > 8$)

If $q \neq 8$, then $\sqrt{q} \notin \mathbb{Q}(\sqrt{2})$, so $[K : \mathbb{Q}] = 4$ and $1, \sqrt{q}, \sqrt{2}, \sqrt{2}\sqrt{q}$ are linearly independent over $\mathbb{Q}$. Their $\mathbb{Z}$-span M is closed under multiplication, because $(\sqrt{q})^2 = q$, $(\sqrt{2})^2 = 2$ and $(\sqrt{2}\sqrt{q})^2 = 2q$ all lie in $\mathbb{Z}$, and the cross products stay in the span. M contains 1, $\sqrt{q}$ and $\sqrt{2}$. So $M = \mathbb{Z}[\sqrt{q}, \sqrt{2}] = \mathcal{O}$, and the four elements form a $\mathbb{Z}$-basis. All four are algebraic integers, so $\mathcal{O}$ lies in the ring of integers of K.

C.3 The Frobenius congruences and (3), (4) (fills B7, B8)

Let p be odd with $p \nmid q$. Then $(\sqrt{q})^p = q^{(p-1)/2}\sqrt{q}$. Euler's criterion gives $q^{(p-1)/2} \equiv (q/p) \pmod p$, and for odd $p \nmid q$ the Kronecker symbol equals the Legendre symbol, so $(q/p) = \chi(p)$. Hence $(\sqrt{q})^p - \chi(p)\sqrt{q} = (q^{(p-1)/2} - \chi(p))\sqrt{q} \in p\mathbb{Z}\cdot\sqrt{q} \subset p\mathcal{O}$. The same argument with 2 in place of q gives $(\sqrt{2})^p \equiv (2/p)\sqrt{2}$. Then $(\sqrt{2}\sqrt{q})^p \equiv (2/p)\chi(p)\sqrt{2}\sqrt{q}$.

In the commutative ring $\mathcal{O}$, $(a + b + c + d)^p \equiv a^p + b^p + c^p + d^p \pmod{p\mathcal{O}}$, because every multinomial coefficient other than the extreme ones is divisible by p. Also $n^p \equiv n \pmod p$ for integers n. So

$$\theta_n^p \equiv n_1 + n_2\chi(p)\sqrt{q} + n_3(2/p)\sqrt{2} + n_4(2/p)\chi(p)\sqrt{(2q)} \pmod{p\mathcal{O}}.$$

With $\chi(p) = -1$: - If $(2/p) = 1$, this is $n_1 - n_2\sqrt{q} + n_3\sqrt{2} - n_4\sqrt{(2q)} = \sigma\theta_n$. - If $(2/p) = -1$, this is $n_1 - n_2\sqrt{q} - n_3\sqrt{2} + n_4\sqrt{(2q)} = \sigma\tau\theta_n$, since $\sigma\tau$ sends $\sqrt{q} \mapsto -\sqrt{q}$, $\sqrt{2} \mapsto -\sqrt{2}$ and $\sqrt{(2q)} \mapsto \sqrt{(2q)}$.

∎

C.4 Lemma 2 (fills B9)

Write $i = pe + v$ with $0 \leq v < p$. Then

$$G_{i,j,k} = \Sigma_{t=0}^{e} C(e, t)(-1)^t x_1^{i-pt} x_\ell^t x_2^j x_3^k.$$

The monomial for t has weighted degree $(i - pt) + 8(j + k + t) = i + 8(j + k) - t(p - 8) \leq D$, because $p > 8$. So $G_{i,j,k} \in$

L_D. The t = 0 term is $x_1^i x_2^j x_3^k$, with coefficient 1. Every other term has x_1 exponent $i - pt < i$.

Order the index set by decreasing x_1 exponent, breaking ties arbitrarily. In that order the matrix expressing the G's in terms of the monomials is triangular with ones on the diagonal and integer entries. Its inverse is also integral and unitriangular, so the G's form an O-basis of L_D. ∎

C.5 Lemma 3 norm bound (fills B10, B11)

Let $U = \lfloor H/(8\sqrt{|q|}B^{i+j+k})\rfloor$. For $0 \le u_t \le U$, each embedding of $c = u_1 + u_2\sqrt{q} + u_3\sqrt{2} + u_4\sqrt{(2q)}$ has absolute value at most $U(1 + \sqrt{|q|} + \sqrt{2} + \sqrt{2}\sqrt{|q|}) = U(1 + \sqrt{2})(1 + \sqrt{|q|})$. For $|q| \ge 3$, $1 + \sqrt{|q|} \le (1 + 1/\sqrt{3})\sqrt{|q|}$, so $(1 + \sqrt{2})(1 + \sqrt{|q|}) \le 3.81\sqrt{|q|} < 8\sqrt{|q|}$. Hence $\|c\|B^{i+j+k} \le 8\sqrt{|q|}\, U\, B^{i+j+k} \le H$.

Distinct $(u_1, \ldots, u_4)$ give distinct c by C.2. Distinct coefficient tuples give distinct f. Finally, $(U + 1)^4 > x^4$ where $x = H/(8\sqrt{|q|}B^{i+j+k})$. ∎

C.6 Lemma 4 details (fills B12 to B15)

Take ℓ as stated. By (3) or (4), $x_1^p - x_\ell$ evaluated at $(\theta_n, \sigma\theta_n, \sigma\tau\theta_n)$ lies in pO. Its e-th power lies in p^eO. The other factors lie in O, because σ and τ map O to O. So $G_{i,j,k}(\text{point}) \in p^{\lfloor i/p\rfloor}O$.

Since $i \le D$, $\lfloor i/p\rfloor \le \lfloor D/p\rfloor \le k_p$. Changing a coordinate c by an element of $p^{k_p - \lfloor i/p\rfloor}O$ therefore changes f(point) by an element of $p^{k_p}O$.

$O \cong Z^4$ as a group, so $|O/mO| = m^4$. Since $P = \Pi\, p^{k_p}$ with distinct primes, the Chinese remainder theorem for Z, tensored with the free Z-module O, gives $O/PO \cong \Pi_p\, O/p^{k_p}O$. The map sending f to its vector modulo PO factors injectively through the product of its vectors modulo each $p^{k_p}O$. So the count is at most the product of the counts. ∎

C.7 The pigeonhole inequality (fills B16 to B19)

There are more polynomials than residue vectors if $(P/(32r\sqrt{|q|}))^{4r}\, B^{-4W} \ge P^{4r}\, \Pi_p\, p^{-4E_p}$. Take logarithms and divide by 4. Then $4r \log P$ cancels, and the condition becomes $\Sigma_p E_p \log p \ge W \log B + r \log(32r\sqrt{|q|})$. This is (11), with ≥ in place of >. The strict inequality in Lemma 3 makes either form enough.

The remaining bounds: - $\|F\|_B \le 2H$, by the triangle inequality in each embedding. - $\|\theta_n\| \le (N - 1)(1 + \sqrt{2})(1 + \sqrt{|q|}) < B$, by C.5. - $\|\sigma\theta_n\| = \|\sigma\tau\theta_n\| = \|\theta_n\|$, because σ and τ permute the four embeddings. - In each embedding, $|F(\text{point})| \le \Sigma\, \|c\|B^{i+j+k} \le r \cdot 2H = P/2$. - For $0 \ne \alpha = P\beta$ with $\beta \in O$: $N(\alpha) = P^4 N(\beta)$, with $N(\beta) \in Z \setminus \{0\}$ because β is a nonzero algebraic integer. So $|N(\alpha)| \ge P^4$. But $|N(\alpha)| = \Pi_t\, |\alpha^{(t)}| \le (P/2)^4$. Contradiction, so $\alpha = 0$.

∎

C.8 The asymptotics and "(9) implies (11)" (fills B20, B21)

Let $R_D = \{(i, j, k) \in Z^3_{\ge 0} : i + 8(j + k) \le D\}$. Put $s = j + k$. The number of pairs (j, k) with $j + k = s$ is $s + 1$, and for each s, i runs over $0, \ldots, D - 8s$.

The continuous analogues, with area element s ds for (j, k): - **r:** $\int_0^{D/8} s(D - 8s)\, ds = D^3/128 - D^3/192 = \mathbf{D^3/384}$. - **Σ i:** $\int_0^{D/8} s(D - 8s)^2/2\, ds = \mathbf{D^4/1536}$. - **Σ (j + k):** $\int_0^{D/8} s^2(D - 8s)\, ds = D^4/1536 - D^4/2048 = \mathbf{D^4/6144}$. - **W = Σ i + Σ (j + k):** $4D^4/6144 + D^4/6144 = \mathbf{5D^4/6144}$.

All by hand. The lattice sums differ from these integrals by $O(D^2)$ for r and $O(D^3)$ for the degree-one sums. Each is a sum of a polynomial of degree $d \le 1$ over the lattice points of the dilate $D \cdot R_1$ of a fixed polytope, and the error is bounded by the boundary contribution, $O(D^{d+2})$. Step 3 checks these exactly.

E_p. Since $0 \le i/p - \lfloor i/p\rfloor < 1$, $E_p = (1/p)\Sigma\, i - \theta_p\, r$ with $0 \le \theta_p < 1$. So $E_p = D^4/(1536p) + O(D^3)$, uniformly in p.

(9) implies (11). Multiply E_p by $\log p$ and sum over S. The error is $\ll D^3\, \Sigma_{p \le Q} \log p \ll D^3 Q$ by Chebyshev. So

$\Sigma\, E_p \log p \ge (D^4/1536)(13/10) \log B - C\, D^3 Q$.

The right side of (11) is at most

$(D^4/1536)(5/4) \log B + C\, D^3 \log B + C\, D^3 \log(32r\sqrt{|q|})$.

The difference is at least

$(D^4/1536)(1/20) \log B - C\, D^3 Q - C\, D^3 \log B - C\, D^3 \log(32r\sqrt{|q|})$.

Since $B \ge N$, the first term is $\ge D^4 \log N / 30720$. Against it: - $D^3 Q = D^4/(\log N)^2$. - $\log B \le 2 \log N + 3$. - $\log(32r\sqrt{|q|}) \ll \log D + \log|q| \ll \log N$, because $D \le N^{4/3}$ and $|q| \le N$ for $\gamma \ge 1$.

Every error term is $o(D^4 \log N)$ with constants that do not involve γ. So (11) holds for $N \geq N_0$, with N_0 absolute. ∎

C.9 Mertens and Lemma 1, and the decisive inequality (fills B26 to B28)

Order of choices. Let C_1 be the constant in Lemma 1. Fix the integer $\gamma \geq 90(C_1 + 2)$. Then fix $\varepsilon = 1/(160\, C_1\, \gamma)$. Then take $|q| \geq q_0(\gamma)$, large enough for every "N large" condition below. Suppose $(1 - \beta)\log|q| < \varepsilon$. Put $\mathscr{L} = \log|q|$, so $\log N = \gamma\mathscr{L}$.

Lower bound for the sum over S. Mertens gives $\Sigma_{p \leq Q} (\log p)/p \geq \log Q - C_M$. The primes $p \leq Q$ not in S fall into three groups: - $p \leq 8$: they contribute less than 2. - $p \mid q$ with $p > 8$: they contribute at most $\Sigma_{p|q} \log p \leq \mathscr{L}$. - $\chi(p) = 1$ ($p \nmid 2q$): by Lemma 1 with $X = Q \geq 3$, they contribute at most $C_1(\mathscr{L} + (1 - \beta)(\log Q)^2)$.

Since $Q \leq D < N^{4/3}$, $(\log Q)^2 < (16/9)\gamma^2\mathscr{L}^2$. So $(1 - \beta)(\log Q)^2 < (16/9)\varepsilon\gamma^2\mathscr{L} = (16/9)\varepsilon\gamma \log N$.

From $D \geq \tfrac{1}{2}((N - 1)/3)^{4/3} - 1$, $\log Q \geq (4/3)\log N - 2 \log\log N - C$. Therefore

$\Sigma_{p \in S} (\log p)/p \geq \log N\, [4/3 - (1 + C_1)/\gamma - (16/9)C_1\varepsilon\gamma] - 2 \log\log N - C'$.

The target. $(13/10) \log B = \log N\, [13/10 + 13/(20\gamma)] + (13/10)\log 8$.

Comparison. (9) holds when

$\log N\, [1/30 - (1 + C_1 + 13/20)/\gamma - (16/9)C_1\varepsilon\gamma] > 2 \log\log N + C''$.

With the choices above: - $(1 + C_1 + 13/20)/\gamma \leq 1/90$. - $(16/9)C_1\varepsilon\gamma = 16/1440 = 1/90$.

So the bracket is at least $1/30 - 2/90 = 1/90$. The inequality then holds for N large, that is, for $|q| \geq q_0(\gamma)$. ∎

C.10 The vectors, the determinant and the kernel (fills B29 to B32)

Fix an embedding $K \hookrightarrow \mathbb{C}$, with $\sqrt{(2q)} := \sqrt{2}\sqrt{q}$. Write $a = \sqrt{q}$ and $b = \sqrt{2}$. By the definitions of σ and $\sigma\tau$, $(\theta_n, \sigma\theta_n, \sigma\tau\theta_n) = n_1\gamma_1 + n_2\gamma_2 + n_3\gamma_3 + n_4\gamma_4$, with $\gamma_1 = (1, 1, 1)$, $\gamma_2 = (a, -a, -a)$, $\gamma_3 = (b, b, -b)$ and $\gamma_4 = (ab, -ab, ab)$.

Determinant. $\det[\gamma_1; \gamma_2; \gamma_3] = 1\cdot(ab + ab) - 1\cdot(-ab + ab) + 1\cdot(ab + ab) = 4ab = 4\sqrt{(2q)} \neq 0$. So π has rank 3, and $\ker \pi$ has dimension 1.

Kernel. $\pi(v) = ab\cdot\gamma_1 + b\cdot\gamma_2 - a\cdot\gamma_3 - \gamma_4$. Coordinate by coordinate: - $ab + ab - ab - ab = 0$; - $ab - ab - ab + ab = 0$; - $ab - ab + ab - ab = 0$.

So $\ker \pi = \mathbb{C}v$.

No rational subspace contains ker π. If $\ker \pi$ lay in a proper subspace spanned by rational vectors, that subspace would lie in a hyperplane $\Sigma\, a_i x_i = 0$ with $a \in \mathbb{Q}^4 \setminus \{0\}$, and then $\Sigma\, a_i v_i = 0$. That contradicts the Q-independence of $1, \sqrt{q}, \sqrt{2}, \sqrt{(2q)}$, which holds for $q \neq 8$ by C.2.

Final contradiction. $F \in L_D$ has total degree at most D, because $i + j + k \leq i + 8(j + k) \leq D$. Its image under the embedding is a nonzero complex polynomial of the same degree. With $M = N - 1 \geq 3$ and $\mu \geq 4/3$, $(M/3)^{\mu} \geq (M/3)^{4/3} > D$. Since rank $\Gamma = 4$, $\gamma_1, \ldots, \gamma_4$ is a basis of Γ, which is the hypothesis Masser's Theorem 1 needs (Step 4 to confirm). ∎

C.11 Small |q| (fills B2)

For each of the finitely many fundamental discriminants with $|q| < q_0(\gamma)$, $L(s, \chi)$ is entire and not identically zero, so its zeros in $[0, 1]$ are finite in number (F14). Also $L(1, \chi) \neq 0$ (F14), so every real zero satisfies $\beta < 1$. Hence c_small := min over these q and their real zeros in $(0, 1)$ of $(1 - \beta)\log|q|$ is positive, or there are no such zeros. Take $c = \min(\varepsilon, c_small)$.

Whether c can be computed (the effectivity question) depends on whether C_1, C_M, the Chebyshev constant and N_0 are explicit, and on a finite computation for small q. That question is reserved for Step 5. ∎

Proposed Step 3: machine recomputation

In Python, with exact rational arithmetic (sympy and fractions), saved as scripts in your folder alongside the report:

1. **Lattice sums.** Exact r(D), W(D) and Σ i for D up to several thousand, compared with $D^3/384$, $5D^4/6144$ and $D^4/1536$, with the observed error exponents. Then E_p against $D^4/(1536p)$, for many p.
2. **Algebra.** Symbolic: $\det(\gamma_1, \gamma_2, \gamma_3) = 4\sqrt{(2q)}$, $\pi(v) = 0$, and the identity $(\theta_n, \sigma\theta_n, \sigma\tau\theta_n) = \Sigma\, n_j\gamma_j$. Numerical, for sample q: the Frobenius congruences (3) and (4) in O, for many primes p.
3. **Lemma 2.** For small p and D: the G's lie in L_D, and the change-of-basis matrix is unitriangular.
4. **Lemma 3 constant.** max over q of $(1 + \sqrt{2})(1 + \sqrt{|q|})/\sqrt{|q|} < 8$.
5. **q = 8.** A concrete demonstration of FS1.

6. **Margin chain.** $5/4 < 13/10 < 4/3$. Then, with illustrative constants C_1, the parameter choices of C.9, to show that the order of choices closes.

The output will be "Step 3 - Kerr Landau-Siegel proof evaluation Step 3 18 Oct 2026 HHMM_v1.md" plus the scripts.